

Predict the Branch

Instructor Answer Key | CISP 360 - Lecture 06.02: Conditionals

Instructor use: Have students commit to a prediction before revealing the answer.

Answer Key

Example 1

Output:	Pass
Reasoning:	The condition <code>72 >= 70</code> is true, so the statement inside the if block executes.

Example 2

Output:	Welcome
Reasoning:	The condition <code>17 >= 18</code> is false, so Adult is skipped. Welcome is outside the if statement and always executes.

Example 3

Output:	Not freezing
Reasoning:	The condition <code>32 < 32</code> is false because the comparison is strict. The else branch executes.

Boundary reminder

Example 3 uses `<` rather than `<=`. At exactly 32 degrees the condition is false, so execution goes to else.

Predict the Branch

Instructor Answer Key - Continued

Example 4

Output:	B
Reasoning:	85 >= 90 is false. The next condition, 85 >= 80, is true, so the if/else if chain stops at B.

Example 5

Output:	ABC
Reasoning:	12 > 10 is true, so A prints. Then 12 < 20 is true, so B prints. The final cout is outside both if statements, so C prints.

Debrief Answers

1.	The condition at the current decision point is evaluated first.
2.	Statements inside false branches and later branches in a completed if/else if chain are skipped.
3.	Example 4 stops at B because an if/else if chain executes only the first true branch.
4.	The nested if in Example 5 is tested only after the outer condition is true.
5.	Boundary behavior depends on the operator. For example, < excludes equality while >= includes it.